

INSIDER ONE

Software Engineer Assessment

Project: Event-Driven Notification System

Overview

Design and implement a scalable notification system that processes and delivers messages through multiple channels (SMS, Email, Push). The system should handle high throughput, ensure reliable delivery, and provide real-time status tracking.

Technology Choice: You may implement this project using **Go (1.2x)**.

Business Context

Insider One needs to send millions of notifications daily to users across different channels. The system must handle burst traffic (e.g., flash sales, breaking news), retry failed deliveries intelligently, and provide visibility into delivery status for both internal teams and API consumers.

Functional Requirements

1. Notification Management API

- Create notification requests with recipient, channel, content, and priority
- Support batch creation (up to 1000 notifications per request)
- Query notification status by ID or batch ID
- Cancel pending notifications
- List notifications with filtering (status, channel, date range) and pagination

2. Processing Engine

- Process notifications asynchronously via queue workers
- Implement rate limiting: maximum 100 messages per second per channel
- Priority queue support (high, normal, low)
- Content validation (character limits, required fields)
- Idempotency support to prevent duplicate sends

3. Delivery & Retry Logic

- Will thinking and design by candidate

4. Observability

- Real-time metrics endpoint (queue depth, success/failure rates, latency)
- Structured logging with correlation IDs
- Health check endpoint

External Provider Integration

Use **webhook.site** to simulate the external notification provider. Create your own webhook URL and configure the response:

Request Format:

```
POST https://webhook.site/{your-uuid}
```

```
{ "to": "+905551234567", "channel": "sms", "content": "Your message" }
```

Expected Response (Status: 202):

```
{ "messageId": "uuid-here", "status": "accepted", "timestamp": "ISO8601" }
```

Bonus Features (Optional)

Implementing these features demonstrates advanced skills and will be viewed favorably:

- **Failure Handling**
- **Scheduled Notifications:** Allow scheduling notifications for future delivery
- **Template System:** Support message templates with variable substitution
- **WebSocket Updates:** Real-time status updates via WebSocket
- **Distributed Tracing**
- **GitHub Actions CI/CD:** Automated testing and linting pipeline

Deliverables

1. **Source Code:** GitHub/GitLab repository with clean commit history
2. **README.md:** Setup instructions, architecture overview, API examples
3. **Docker Compose:** One-command setup (docker-compose up)
4. **API Documentation:** Swagger/OpenAPI specification
5. **Database Migrations:** Versioned schema changes
6. **Test Suite:** Runnable with single command

Timeline

Expected Duration: 12 hours

Please submit your solution by sharing the repository link. We understand you have other commitments, so if you need additional time, just let us know.

Questions?

If you have any questions about the requirements or need clarification, please don't hesitate to reach out. We're happy to help!

Good luck! We look forward to reviewing your solution.